SwegHammer

A simulation-driven approach to balancing Warhammer 40,000

Project handout — June 2026

What is Warhammer 40,000?

Warhammer 40,000 (often shortened to 40K) is a tabletop miniatures game set in a far-future science-fiction universe. Two players assemble armies of detailed plastic miniatures, deploy them on a terrain-covered table, and fight a battle over five rounds using dice rolls, measuring tapes, and thick rulebooks.

The game is published by Games Workshop and is currently in its 10th Edition. If you want a feel for the basic rules, the quick-start guide is freely available online:

https://wahapedia.ru/wh40k10ed/the-rules/quick-start-guide/

Datasheets: a unit's identity card

Every unit in the game has a **datasheet** — a page that lists all of its stats, weapons, and special abilities. Two example datasheets are reproduced at the end of this handout (pages 5–6). They illustrate the range of complexity in the game: a basic infantry squad (*Intercessor Squad*) and a towering war machine (*Knight Crusader*).

Each model in the game has a core stat line:

Stat	**What it means**
Movement (M)	How far the model moves each turn, in inches
Toughness (T)	How hard the model is to wound
Save (Sv)	Armour save — roll this or higher on a d6 to block a hit
Wounds (W)	Hit points; the model is destroyed when these reach zero
Leadership (Ld)	Morale; lower is better
Objective Control (OC)	How much weight the model carries when holding an objective

Weapons have their own profiles: number of Attacks, Ballistic Skill (the roll needed to hit), Strength, Armour Penetration, and Damage.

The balance problem

Warhammer 40,000 is a grossly complex game, with hundreds of unique models, special abilities, and strategies. Creating a "balanced game" is incredibly tough. When two opponents play a game, a pre-set points limit is agreed (usually 2,000 for a tournament). Each model, character, tank, and monster has an in-game points cost. Players build their army up to that limit, so points are the currency that is supposed to make unlike things comparable: five cheap infantry squads should, in theory, put up a fair fight against one expensive super-heavy walker.

These points costs are set by Games Workshop, who own Warhammer. Sometimes these costs are massively over- or under-tuned, leading to tournaments where players "spam" the best units. This introduces a lot of "feel bad moments" and means older models tend to get shelved to gather dust. Players are often forced to buy models they do not really want in order to keep up with the competitive meta.

The aim of SwegHammer is to eliminate these feel-bad moments and attempt to balance the game with a mathematically and simulation-driven approach to deriving points costs.

How the simulator works

At its core, SwegHammer is a stochastic combat simulator that plays out thousands of 40K battles automatically and measures which units win, which lose, and by how much. It implements the full 10th Edition rules:

- All five combat phases: Command, Movement, Shooting, Charge, Fight (plus Battleshock at round end).
- Real dice rolls for the complete hit → wound → save → feel-no-pain chain — no averaging, so the natural variance of the game is preserved.
- A 2D map with terrain: line-of-sight checks, cover, ruins, and objective markers.
- Around 1,500 units from the full 10th Edition catalogue, including faction rules, detachments, leader auras, and stratagems.

Measuring balance

The simulator runs many thousands of matched battles. If a unit is correctly costed, it should win roughly 50% of its games when fielded at equal points against a representative spread of opponents. A unit that wins significantly more than 50% is under-costed (too cheap for what it does); a unit that wins significantly less is over-costed (too expensive).

By measuring this win rate across the entire catalogue, SwegHammer can identify which units are too strong, which are too weak, and by how much — then adjust their points costs to bring everything closer to an even 50%.

The maths: Lanchester's Square Law

The theoretical backbone comes from **Lanchester's Square Law**, a model from military operations research. In aimed-fire combat (where each combatant picks a specific target), combat power scales with the *square* of the number of combatants:

$$\alpha \cdot A^2 \; - \; \beta \cdot B^2 \; = \; \text{constant}$$

where A and B are the surviving unit counts on each side, and α and β are per-unit combat effectiveness ratings. Side A wins when $\alpha \cdot A_0^2 > \beta \cdot B_0^2$.

This means **numbers matter more than individual quality** — doubling the size of your army quadruples your aggregate combat power. The points system has to account for this: cheap, numerous units need to be priced so that their squared-count advantage is paid for.

SwegHammer defines a unit's raw effectiveness as:

$$e = \text{health} \times \text{damage} \times \text{hit probability}$$

and anchors the entire cost scale to a single baseline unit: one Space Marine with a bolter ($e \approx 0.667$, costed at 15 points). Every other unit's price is derived relative to this anchor, initially through a linear formula, then refined by empirical simulation data until the win-rate spread across the whole catalogue flattens toward 50%.

Why alternating activations matter

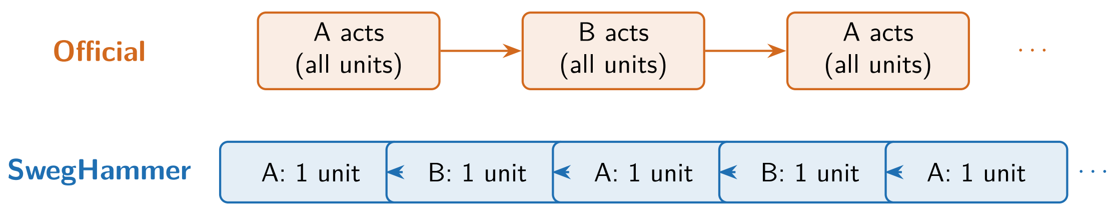
In the official rules, Warhammer 40K uses an "I-Go-You-Go" turn structure: Player A completes their *entire* turn — moving, shooting, and fighting with every unit — before Player B gets to do anything. This creates a massive **first-player advantage**: the player who goes first gets an "alpha strike", an entire turn of unopposed fire before the opponent can respond. Tournament data shows this yields roughly a 58% win rate for the first player and 42% for the second — a gap too large to ignore when trying to balance unit costs.

SwegHammer's solution: unit-by-unit activation

Instead of one player acting with their whole army, SwegHammer uses **alternating activations**: players take turns activating a single unit at a time. Player A activates one unit (it moves, shoots, charges, fights); then Player B activates one of theirs; then back to Player A, and so on until both armies have activated everything.

This dramatically reduces the first-player advantage. The remaining edge (one unit acts before its counterpart) is negligible compared to the official system's full-army-turn head start. Both sides take casualties concurrently, which restores the continuous-fire assumption that Lanchester's model depends on.

Alternating activations also create more interesting tactical decisions: the order in which you activate your units matters, because your opponent gets to react between each one. It rewards planning and adaptation rather than simply going first with the biggest gun.



The two-stage pipeline

SwegHammer's work is split into two sequential stages:

Stage 1 — Make the simulator play like reality. Run the simulator, compare per-faction win rates against real tournament data (a 10,000-game aggregate from the Warp Friends community), and tweak the simulator's rules until the simulated results match the real-world results. The target is a mean absolute error of 2.0 percentage points or less per faction. This stage is about *simulator fidelity*: if the sim does not play like the real game, any prices it produces are meaningless.

Stage 2 — Fit the points equation. Once the simulator is faithful, freeze its rules and fit one master equation that maps every unit's stats to a points cost, plus small per-unit corrections for edge cases. Run the now-trustworthy simulator with equation-priced units, measure the win-rate spread, adjust the equation's coefficients, and repeat until every unit hovers near 50% in equal-points fights. The output is a single formula that prices any unit from its stat line — not a hand-tuned table, but a principled, reproducible result.

Current status and what comes next

Stage 1 is in active development. The simulator's mean absolute error against tournament data currently sits at around 7 percentage points — meaningful progress, but still above the 2-point target. Work is focused on improving the fidelity of faction-specific rules, terrain interactions, and the strategy layer that governs how simulated units choose their actions.

Stage 2 tooling is built and running in parallel, but its outputs are provisional until Stage 1 converges. Once the simulator plays like the real game, the points equation will be re-fitted against the faithful simulation, producing the final balanced points costs.

The long-term vision is a community tool where players can look up simulation-derived points for any unit, explore what-if scenarios ("what if this unit had one more wound?"), and play games with SwegHammer-balanced army lists.

Example datasheets

The two datasheets below illustrate the extremes of the game's unit spectrum. The *Intercessor Squad* is a bread-and-butter infantry squad — cheap, numerous, and straightforward. The *Knight Crusader* is a towering war machine that costs almost five times as much and carries an arsenal of heavy weapons. The points system must make a game between five Intercessor squads and one Knight Crusader feel fair.

Full datasheets with special abilities and stratagems are available on Wahapedia (links below each table).

Intercessor Squad (Space Marines)

M	T	Sv	W	Ld	OC
6"	4	3+	2	6+	2

Ranged weapons

Weapon	Range	A	BS	S	AP	D	Keywords
Bolt rifle	24"	2	3+	4	−1	1	Assault, Heavy
Bolt pistol	12"	1	3+	4	0	1	Pistol
Astartes grenade launcher (krak)	18"	1	3+	9	−2	D3	—

Melee weapons

Weapon	A	WS	S	AP	D
Close combat weapon	3	3+	4	0	1
Power weapon (Sergeant)	4	3+	5	−2	1
Power fist (Sergeant)	3	3+	8	−2	2

Unit size: 5–10 models
Points: 80 (Games Workshop)
Keywords: Infantry, Battleline

Full datasheet: https://wahapedia.ru/wh40k10ed/factions/space-marines/Intercessor-Squad

Knight Crusader (Imperial Knights)

M	T	Sv	W	Ld	OC
10"	12	3+	22	6+	10

Invulnerable save: 5+ (against ranged attacks; 5+ in melee via Ion Shield)

Ranged weapons

Weapon	Range	A	BS	S	AP	D	Keywords
Avenger gatling cannon	36"	18	3+	6	−2	2	—
Thermal cannon	24"	2D3	3+	12	−4	D6	Blast, Melta 2
Rapid-fire battle cannon	72"	2D6+6	3+	10	−1	3	Blast
Ironstorm missile pod	48"	D6+1	3+	5	0	1	Blast, Indirect Fire
Stormspear rocket pod	48"	3	3+	8	−2	D6	—
Meltagun	12"	1	3+	9	−4	D6	Melta 2

Melee weapons

Weapon	A	WS	S	AP	D
Titanic feet	4	4+	8	−1	2

Unit size: 1 model
Points: 385 (Games Workshop)
Keywords: Vehicle, Walker, Titanic, Towering, Character
Deadly Demise: D6 (when destroyed, nearby units suffer D6 mortal wounds)

Damaged profile: When this model has 1–7 wounds remaining, subtract 1 from its hit rolls (both ranged and melee), and halve its Move characteristic.

Full datasheet: https://wahapedia.ru/wh40k10ed/factions/imperial-knights/Knight-Crusader